

Facet Joint Injection

ACG: A-0695 (AC)

[Link to Codes](#)

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Clinical Indications

- Facet joint injection may be indicated when **ALL** of the following are present(1):
 - Diagnostic block (medial branch nerve block, intra-articular injection) needed to confirm facet joint as source of spinal pain, as indicated by **1 or more** of the following(5)(6)(7)(19)(20)(21):[\[A\]](#)
 - Initial diagnostic block to diagnose facet pain (dual diagnostic blocks are necessary to diagnose facet pain)
 - Second confirmatory diagnostic block (dual diagnostic blocks are necessary to diagnose facet pain) if documentation indicates first diagnostic block produced 80% or greater relief of primary (index) pain and duration of relief is consistent with agent employed
 - Patient is candidate for facet neurotomy,^[A] as indicated by **ALL** of the following(2)(3):
 - Chronic spinal pain (at least 3 months' duration) originating from **1 or more** of the following:
 - Cervical spine (eg, following whiplash injury)(5)
 - Lumbar spine(25)
 - Failure of nonoperative management, as indicated by **1 or more** of the following(26):
 - Exercise program
 - Pharmacotherapy^[B]
 - Physical therapy or spinal manipulation therapy
 - Imaging studies and physical examination have ruled out other causes of spinal pain (eg, herniated disk, spinal stenosis, fracture, tumor).
 - No coagulopathy(28)
 - No current infection

Alternatives

- Nonoperative management may include(1)(2)(29):
 - Cognitive behavioral therapy
 - Exercise program
 - Pharmacotherapy(30)
 - Physical therapy. See Spine Soft Tissue Dysfunction Rehabilitation [\[C\]](#) AC for further information.
 - Spinal manipulation therapy. See Spinal Manipulation Therapy (SMT), Chiropractic and Other [\[C\]](#) AC for further information.

Evidence Summary

Background

A potential source of spinal pain is the posterior zygapophysial joint (facet, Z joint), which adjoins adjacent vertebrae and is innervated by medial branches of the dorsal spinal nerves at 2 levels; however, there is no single history or physical examination finding considered pathognomonic for Z joint syndrome.(2)(3) **(EG 2)**

Fluoroscopic-guided diagnostic medial branch block or intra-articular injection is utilized to identify patients with facet joint pain who are potential candidates for radiofrequency neurotomy. Intra-articular injection is more difficult to perform, and it may have lower prognostic power to determine which patients will respond to neurotomy.(2)(3)(4) **(EG 2)**

Diagnosis of facet joint pain can be made when controlled local anesthetic blockade of the medial branches of the posterior rami of the spinal nerves that supply the putative painful joint(s), or anesthetic delivered via intra-articular facet joint injection, provides relief of the target pain. In controlled diagnostic testing, the patient receives an injection of a short-acting anesthetic agent (lidocaine), and those patients who have at least 80% short-term pain reduction from baseline pain scores are then injected with a longer-acting agent (bupivacaine). Patients with at least 80% pain reduction from baseline pain scores after injection of the longer-acting anesthetic agent are considered to be candidates for facet neurotomy.(1)(5)(6)(7) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 26 published peer reviewed articles, 1 specialty society or other evidence-based guideline, and 1 Cochrane systematic review.

For diagnostic block, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** The goal of facet joint injection is to make a diagnosis of facet joint pain to determine if a patient is a candidate for facet neurotomy. In controlled diagnostic testing, the patient receives an injection of a short-acting anesthetic agent (eg, lidocaine); those patients who have at least 80% short-term pain reduction from baseline pain scores are then injected with a longer-acting agent (bupivacaine). Patients with at least 80% pain reduction from baseline pain scores after injection of the longer-acting anesthetic agent are candidates for facet neurotomy. Diagnostic injection of the local anesthetic agent is either performed via blockade of the medial branches of the posterior rami of the spinal nerves that supply the putative painful joint(s), or alternatively via intra-articular injection; medial branch block is associated with a better prediction of success of neurotomy compared with intra-articular injection.(1)(4)(22) **(EG 2)** For diagnosis of cervical facet joint pain, a systematic review of 10 studies (1192 patients) evaluating the diagnostic accuracy of facet joint injections (using 75% or more pain relief as a criterion standard) found that there was good evidence to support their use and that the false-positive rate was 27% to 63%.(23) **(EG 1)** For diagnosis of lumbar facet joint pain, a systematic review of 14 studies (2804 patients) evaluating the diagnostic accuracy of facet joint injections (using 75% or more pain relief as a criterion standard) found that there was good evidence to support their use and that the false-positive rate was 17% to 49%.(23) **(EG 1)** An evidence-based guideline states that there is moderate evidence that the diagnosis of lumbar facet-mediated back pain can be established with the use of a medial nerve double-injection technique that results in a greater than 80% pain improvement threshold; the results are predictive of a good response to facet neurotomy.(13) **(EG 2)** A multispecialty consensus practice guideline recommends that a 50% or more reduction in pain be used to define a positive diagnostic block, acknowledging that the exact amount of pain reduction for a positive block remains uncertain; further studies are needed to identify optimal response criteria.(2) **(EG 2)** For diagnosis of thoracic facet joint pain, a systematic review of 3 studies from the same research group (183 patients) evaluating the diagnostic accuracy of facet joint injections (using 80% or more pain relief as a criterion standard) found that there was good evidence for their use and that the false-positive rate was 42% to 58%.(23) **(EG 1)**

Inconclusive or Non-Supportive Evidence

For therapeutic intra-articular facet joint injection, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** An evidence-based technology assessment found limited evidence to suggest that facet joint corticosteroid injections are not effective for presumed facet joint pain.(8) **(EG 1)** A systematic review of cervical intra-articular facet joint injection identified only 2 randomized trials that reported conflicting outcomes (one reported negative outcomes and the other reported indeterminate results); these studies had significant limitations, which included lack of placebo controls, confounding variables of trigger point and botulinum toxin injection, and a withdrawal rate of greater than 20%. The authors concluded that the evidence supporting cervical intra-articular facet joint injection was of moderate or low quality.(9) **(EG 1)** For lumbar intra-articular facet joint injection, a systematic review of 5 randomized controlled trials found that the evidence supporting its use was of moderate or low quality.(9) **(EG 1)** A systematic review of 6 randomized controlled trials (434 patients) evaluating the efficacy of therapeutic lumbar intra-articular facet joint injections with active drug found that there is insufficient high-quality evidence to support their use over a sham procedure, placebo injection, or conservative therapy.(10) **(EG 1)** A randomized double-blind controlled study of 28 patients with lumbar Z joint pain confirmed by medial branch block comparing intra-articular fluoroscopic-guided corticosteroid injection with saline injection found no significant difference in the average time to, or percentage of patients receiving, subsequent radiofrequency ablation. The study was terminated early because more than 75% of patients in both groups underwent radiofrequency ablation by the first study follow-up visit at 6 weeks. The authors advise larger, adequately powered analyses.(11) **(EG 1)** A randomized trial of 32 patients with lumbar facet joint pain compared treatment with either lumbar medial branch radiofrequency ablation or intra-articular facet corticosteroid injection and found, at 3-month follow-up, that patients who received radiofrequency

ablation were more likely to report 50% or greater improvement in numerical pain score than those who received corticosteroid injection. The authors noted that the study was limited by small sample size, lack of blinding, and treatment crossover.(12) **(EG 1)** An evidence-based guideline states that there is moderate evidence suggesting the lack of a role for intra-articular facet joint injections for treatment of facet-mediated chronic low back pain without radiculopathy in patients with degenerative disease of the lumbar spine.(13) **(EG 2)** For thoracic intra-articular facet joint injection, a systematic review revealed that there was no available literature investigating its efficacy.(9) **(EG 1)** Consensus practice guidelines recommend against the routine use of therapeutic intra-articular facet injections but note there may be a role for this intervention for certain populations, such as patients with contraindications to radiofrequency ablation.(2)(3) **(EG 2)**

For therapeutic medial branch nerve block, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** For cervical facet joint pain, a systematic review identified only one randomized controlled trial and one observational study (from the same research group) supporting the short-term and long-term effectiveness of medial branch block; limitations of the studies included lack of a placebo group in the randomized controlled trial and lack of randomization in the observational study.(9) **(EG 1)** A randomized double-blind controlled trial of 120 patients with cervical facet joint pain who received medial branch nerve block with local anesthetic with or without steroids found, at 2-year follow-up, that although both groups had significant improvement in pain scores and functional assessment, there was no significant difference between the groups. In addition, although both groups showed a decrease in opioid intake, the difference was not significant. The authors noted that the study lacked a placebo group and recommended larger placebo-controlled studies to validate the findings.(14) **(EG 1)** For lumbar facet joint pain, a systematic review identified only 2 randomized controlled studies of moderate-quality to good-quality evidence supporting the use of lumbar facet nerve block.(15) **(EG 1)** A randomized double-blind controlled trial of 120 patients with lumbar facet joint pain who received medial branch nerve block with local anesthetic with or without steroids found, at 2-year follow-up, that although both groups had significant improvement in pain scores and functional assessment, there was no significant difference between the groups. In addition, although both groups showed a decrease in opioid intake, the difference was not significant. The authors noted that the study lacked a placebo group and recommended larger placebo-controlled studies to validate the findings.(16) **(EG 1)** For thoracic facet joint pain, a systematic review identified only one randomized trial and one observational study (from the same research group) supporting the use of thoracic facet joint nerve block.(9) **(EG 1)** A randomized double-blind controlled trial of 100 patients with thoracic facet joint pain who received medial branch nerve block with local anesthetic with or without steroids found, at 2-year follow-up, that although both groups had significant improvement in pain scores and functional assessment, there was no significant difference between the groups. In addition, although both groups showed a decrease in opioid intake, the difference was not significant. The authors noted that the study lacked a placebo group and recommended larger placebo-controlled studies to validate the findings.(17) **(EG 1)** For osteoporotic compression fracture, a retrospective study of 53 patients who underwent therapeutic medial branch block found that there was significant improvement in pain and disability scores at 12-month follow-up. The study was limited by performance of a single diagnostic block and lack of a placebo control group. Additional double-blind randomized controlled studies were recommended.(18) **(EG 2)** A consensus practice guideline recommends against the routine use of therapeutic medial branch blocks but notes there may be a role for this intervention for certain populations, such as patients with contraindications to radiofrequency ablation.(2) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(31); 42 CFR 422.101(32)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(33); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(34); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(35)

Medicare Coverage Determinations: Medicare Coverage Database(36)

References

1. Manchikanti L, et al. Comprehensive evidence-based guidelines for facet joint interventions in the management of chronic spinal pain: American Society of Interventional Pain Physicians (ASIPP) guidelines facet joint interventions 2020 guidelines. Pain Physician 2020;23(3S):S1-S127. [Context Link 1, 2, 3, 4, 5] View abstract...

2. Cohen SP, et al. Consensus practice guidelines on interventions for lumbar facet joint pain from a multispecialty, international working group. *Regional Anesthesia and Pain Medicine* 2020;45(6):424-467. DOI: 10.1136/rapm-2019-101243. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
3. Hurley RW, et al. Consensus practice guidelines on interventions for cervical spine (facet) joint pain from a multispecialty international working group. *Pain Medicine* 2021;22(11):2443-2524. DOI: 10.1093/pm/pnab281. [Context Link 1, 2, 3, 4] View abstract...
4. Perolat R, et al. Facet joint syndrome: from diagnosis to interventional management. *Insights Into Imaging* 2018;9(5):773-789. DOI: 10.1007/s13244-018-0638-x. [Context Link 1, 2] View abstract...
5. Engel A, King W, Schneider BJ, Duszynski B, Bogduk N. The effectiveness of cervical medial branch thermal radiofrequency neurotomy stratified by selection criteria: a systematic review of the literature. *Pain Medicine* 2020;21(11):2726-2737. DOI: 10.1093/pm/pnaa219. [Context Link 1, 2, 3] View abstract...
6. Manchikanti L, et al. Low back pain and diagnostic lumbar facet joint nerve blocks: assessment of prevalence, false-positive rates, and a philosophical paradigm shift from an acute to a chronic pain model. *Pain Physician* 2020;23(5):519-530. [Context Link 1, 2] View abstract...
7. Manchikanti L, et al. Assessment of prevalence of cervical facet joint pain with diagnostic cervical medial branch blocks: analysis based on chronic pain model. *Pain Physician* 2020;23(6):531-540. [Context Link 1, 2] View abstract...
8. Chou R, et al. Pain Management Injection Therapies for Low Back Pain. Technology Assessment Report ESIB0813 [Internet] Agency for Healthcare Research and Quality. 2015 July Accessed at: <https://www.ahrq.gov/>. [created 2015; accessed 2024 Sep 30] [Context Link 1] View abstract...
9. Manchikanti L, et al. A systematic review and best evidence synthesis of the effectiveness of therapeutic facet joint interventions in managing chronic spinal pain. *Pain Physician* 2015 Jul-Aug;18(4):E535-E582. [Context Link 1, 2, 3, 4, 5] View abstract...
10. Vekaria R, Bhatt R, Ellard DR, Henschke N, Underwood M, Sandhu H. Intra-articular facet joint injections for low back pain: a systematic review. *European Spine Journal* 2016;25(4):1266-1281. DOI: 10.1007/s00586-016-4455-y. [Context Link 1] View abstract...
11. Kennedy DJ, et al. Corticosteroid injections into lumbar facet joints: a prospective, randomized, double-blind placebo-controlled trial. *American Journal of Physical Medicine and Rehabilitation* 2018;97(10):741-746. DOI: 10.1097/PHM.0000000000000960. [Context Link 1] View abstract...
12. McCormick ZL, et al. A pragmatic randomized prospective trial of cooled radiofrequency ablation of the medial branch nerves versus facet joint injection of corticosteroid for the treatment of lumbar facet syndrome: 12 month outcomes. *Pain Medicine* 2023;24(12):1318-1331. DOI: 10.1093/pm/pnad107. [Context Link 1] View abstract...
13. Watters WC, et al. Guideline update for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 13: injection therapies, low-back pain, and lumbar fusion. *Journal of Neurosurgery: Spine* 2014;21(1):79-90. DOI: 10.3171/2014.4.SPINE14281. [Context Link 1, 2] View abstract...
14. Manchikanti L, Singh V, Falco FJ, Cash KA, Fellows B. Comparative outcomes of a 2-year follow-up of cervical medial branch blocks in management of chronic neck pain: a randomized, double-blind controlled trial. *Pain Physician* 2010;13(5):437-450. [Context Link 1] View abstract...
15. Manchikanti L, Hirsch JA, Falco FJ, Boswell MV. Management of lumbar zygapophysial (facet) joint pain. *World Journal of Orthopedics* 2016;7(5):315-337. DOI: 10.5312/wjo.v7.i5.315. [Context Link 1, 2] View abstract...
16. Manchikanti L, Singh V, Falco FJ, Cash KA, Pampati V. Evaluation of lumbar facet joint nerve blocks in managing chronic low back pain: a randomized, double-blind, controlled trial with a 2-year follow-up. *International Journal of Medical Sciences* 2010;7(3):124-135. [Context Link 1] View abstract...
17. Manchikanti L, Singh V, Falco FJ, Cash KA, Pampati V, Fellows B. The role of thoracic medial branch blocks in managing chronic mid and upper back pain: a randomized, double-blind, active-control trial with a 2-year followup. *Anesthesiology Research and Practice* 2012;2012:585806. DOI: 10.1155/2012/585806. [Context Link 1] View abstract...
18. Park KD, et al. Effect of medial branch block in chronic facet joint pain for osteoporotic compression fracture: one year retrospective study. *Annals of Rehabilitation Medicine* 2013;37(2):191-201. DOI: 10.5535/arm.2013.37.2.191. [Context Link 1] View abstract...
19. Conger A, et al. The effectiveness of radiofrequency ablation of medial branch nerves for chronic lumbar facet joint syndrome in patients selected by guideline-concordant dual comparative medial branch blocks. *Pain Medicine* 2020;21(5):902-909. DOI: 10.1093/pm/pnz248. [Context Link 1] View abstract...
20. Cohen SP, et al. Effectiveness of lumbar facet joint blocks and predictive value before radiofrequency denervation: the Facet Treatment Study (FACTS), a randomized, controlled clinical trial. *Anesthesiology* 2018;129(3):517-535. DOI: 10.1097/ALN.0000000000002274. [Context Link 1] View abstract...
21. Cohen SP, Hooten WM. Advances in the diagnosis and management of neck pain. *British Medical Journal* 2017;358:j3221. [Context Link 1] View abstract...
22. Cohen SP, Huang JH, Brummett C. Facet joint pain--advances in patient selection and treatment. *Nature Reviews. Rheumatology* 2013;9(2):101-116. DOI: 10.1038/nrrheum.2012.198. [Context Link 1] View abstract...
23. Boswell MV, et al. A best-evidence systematic appraisal of the diagnostic accuracy and utility of Facet (Zygapophysial) joint injections in chronic spinal pain. *Pain Physician* 2015 Jul-Aug;18(4):E497-E533. [Context Link 1, 2, 3] View abstract...
24. Falco FJ, et al. Systematic review of the therapeutic effectiveness of cervical facet joint interventions: an update. *Pain Physician* 2012;15(6):E839-E868. [Context Link 1] View abstract...
25. Maas ET, et al. Radiofrequency denervation for chronic low back pain. *Cochrane Database of Systematic Reviews* 2015, Issue 10. Art. No.: CD008572. DOI: 10.1002/14651858.CD008572.pub2. [Context Link 1] View abstract...
26. Popescu A, Lee H. Neck pain and lower back pain. *Medical Clinics of North America* 2020;104(2):279-292. DOI: 10.1016/j.mcna.2019.11.003. [Context Link 1] View abstract...

27. Dowell D, Ragan KR, Jones CM, Baldwin GT, Chou R. CDC Clinical Practice Guideline for Prescribing Opioids for Pain - United States, 2022. MMWR - Recommendations and Reports 2022;71(3):1-95. DOI: 10.15585/mmwr.rr7103a1. [Context Link 1] View abstract...
28. Mazin DA, Sullivan JP. Lumbar and sacral radiofrequency neurotomy. Physical Medicine and Rehabilitation Clinics 2010;21(4):843-850. DOI: 10.1016/j.pmr.2010.06.010. [Context Link 1] View abstract...
29. Kreiner DS, et al. Diagnosis and Treatment of Low Back Pain. Evidence-Based Clinical Guidelines for Multidisciplinary Spine Care [Internet] North American Spine Society. 2020 Aug Accessed at: <https://www.spine.org/>. [accessed 2024 Sep 24] [Context Link 1]
30. Nonopioid drugs for pain. Medical Letter on Drugs and Therapeutics 2018;60(1540):25-32. [Context Link 1] View abstract...
31. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2024 Oct 08] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
32. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2024 Sep 30 [accessed 2024 Oct 08] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
33. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 14 - Medical Devices Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2024 Sep 26] [Context Link 1]
34. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12425 [Internet] Centers for Medicare & Medicaid Services. 2023 Dec 21 Accessed at: <https://www.cms.gov/manuals/>. [accessed 2024 Feb 01] [Context Link 1]
35. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 16 - General Exclusions From Coverage Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2024 Jun 06] [Context Link 1]
36. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: <https://www.cms.gov/medicare-coverage-database/search.aspx?> Updated 2024 [accessed 2024 Sep 26] [Context Link 1]

Footnotes

[A] Following diagnostic medial branch nerve block, a practice guideline recommends cervical or lumbar radiofrequency neurotomy for patients with chronic nonradicular spinal pain of at least 3 months' duration who have failed conservative care, who lack evidence of other spinal pathology (eg, herniated disk), and who experience at least 80% pain relief from baseline with diagnostic local anesthetic blockade of the medial branches of the posterior rami of the spinal nerves that supply the putative painful joint(s).(1) A systematic review of one randomized sham-controlled and 5 observational studies found fair evidence supporting the use of cervical radiofrequency neurotomy for facet joint pain.(24) A systematic review of 9 randomized controlled trials found moderate-quality to high-quality evidence supporting the use of lumbar radiofrequency neurotomy for facet joint pain.(15) [A in Context Link 1]

[B] There is some evidence to support the short-term benefit of opioids for mild to moderate pain, but the evidence for improvement in function is inconsistent. Given that chronic opioid therapy for noncancer pain often begins with acute opioid prescriptions, clinicians should provide the lowest effective dose for the shortest duration necessary (eg, 2 to 3 days) to alleviate pain when giving prescriptions for acute pain; limiting the duration of opioid therapy can minimize the need to taper to prevent withdrawal symptoms at the end of the course of opioids and limit unused opioids.(27) [B in Context Link 1]

Codes

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